

Metropolitan Cancer Institute, Department of Hematology/Oncology

Morbidity & Mortality Conference Summary

Patient: Mark Lewis

MRN: AML018

DOB: 01/06/1957

Date of Death: July 11, 2023

CASE PRESENTATION FOR M&M REVIEW:

Mr. Lewis was a 66-year-old gentleman with therapy-related AML (prior DLBCL treated with R-CHOP 2018-2019) who died 24 months after diagnosis from progressive disease and complications of salvage therapy.

INITIAL PRESENTATION (July 2021): Presented 07/11/21 with pancytopenia during routine oncology follow-up (in remission from DLBCL x2 years). Labs: WBC 2.1K with 28% blasts, Hgb 7.2, Plt 41K. Bone marrow: 45% blasts, complex karyotype with del(5q), -7, +8. NGS: ASXL1 frameshift mutation, TP53 not detected. Classified as adverse risk given therapy-related AML and ASXL1 mutation.

PRIMARY TREATMENT COURSE:

- 07/19/21: Started 7+3 induction (cytarabine + daunorubicin 60mg/m2 given age and prior anthracycline exposure)
- Day 14 marrow: Hypocellular, 8% blasts
- Day 30: CRi achieved (4% blasts, ANC 650, Plt 78K)
- 09/21-12/21: Received 2 cycles intermediate-dose cytarabine consolidation (dose-reduced given prolonged cytopenias)

DISEASE PROGRESSION (January 2023): Routine labs 01/15/23 showed rising blasts (12% circulating). Marrow confirmed relapse with 38% blasts, evolution to more complex karyotype.

SALVAGE ATTEMPTS AND COMPLICATIONS: *February 2023:* Venetoclax + azacitidine x2 cycles - progressive disease

April 2023: proceeded to allogeneic stem cell transplantation from haploidentical donor (brother)

- Day +3: Neutropenic fever, *Stenotrophomonas* bacteremia
- Day +7: Acute respiratory failure requiring intubation
- Day +10: Fungal pneumonia (*Aspergillus fumigatus*)
- Day +14: Renal failure requiring CRRT

TERMINAL EVENT: Despite broad antimicrobial coverage and supportive care, developed refractory septic shock with multiorgan failure. Family meeting 07/09/23 - decision for comfort care given refractory AML and irreversible organ failure. Expired 07/11/23 at 09:30 with wife and daughters present.

LEARNING POINTS FOR M&M:

1. Therapy-related AML with ASXL1 mutation carries dismal prognosis
2. Consider early transplant evaluation even in CR1 for adverse risk
3. Salvage therapy in elderly with t-AML rarely successful
4. Earlier palliative care integration may improve end-of-life quality

CAUSE OF DEATH:

- Immediate: Multiorgan failure
- Underlying: Refractory AML, therapy-related

Submitted by: James Patterson, MD
Attending Physician